observations have shown that these tiny temperature differences fill out a particular pattern on the sky, demonstrated by the orderly progression of diamonds in Figure 11.1b. And marvel of marvels, calculations done today, using the inflationary framework, are able to *explain* the pattern of these minuscule temperature variations—variations set down nearly 14 billion years ago—and, to top it off, the key to this explanation involves jitters arising from quantum uncertainty. Wow.

This success has convinced many physicists of the inflationary theory's validity. What is of equal importance, these and other precision astronomical measurements, which have only recently become possible, have allowed cosmology to mature from a field based on speculation and conjecture to one firmly grounded in observation—a coming of age that has inspired many in the field to call our era the golden age of cosmology.

Creating a Universe

With such progress, physicists have been motivated to see how much further inflationary cosmology can go. Can it, for example, resolve the ultimate mystery, encapsulated in Leibniz's question of why there is a universe at all? Well, at least with our current level of understanding, that's asking for too much. Even if a cosmological theory were to make headway on this question, we could ask why that particular theory—its assumptions, ingredients, and equations—was relevant, thus merely pushing the question of origin one step further back. If logic alone somehow required the universe to exist and to be governed by a unique set of laws with unique ingredients, then perhaps we'd have a convincing story. But, to date, that's nothing but a pipe dream.

A related but somewhat less ambitious question, one that has also been asked in various guises through the ages, is: Where did all the mass/energy making up the universe come from? Here, although inflationary cosmology does not provide a complete answer, it has cast the question in an intriguing new light.

To understand how, think of a huge but flexible box filled with many thousands of swarming children, incessantly running and jumping. Imagine that the box is completely impermeable, so no heat or energy can